• **ENGINE—7.3L—"KNOCKING/CACKLE" NOISE COMING FROM ENGINE** • **NOISE—"KNOCKING/CACKLE" NOISE COMING FROM ENGINE—VEHICLES WITH 7.3L ENGINE**	**Article No. 00-22-1**

FORD: 1999-2000 ECONOLINE, SUPER DUTY F SERIES
2000 EXCURSION

This article is being republished in its entirety to update the Powertrain Control Module (PCM) recalibration chart information.

ISSUE

Some vehicles equipped with Powerstroke diesel engines after serial number 0896812 (engine build date), may exhibit a noise above the normal sound of the diesel combustion. This noise may sound like a rod knock, but it is intermittent and it increases in intensity as the engine warms up to operating temperature. The "cackle" noise may by caused by pressure variation within the fuel rail and in virtually all instances is attributed to cylinder #8 (driver side rear). The cackle noise does not damage the engine, or decrease the engine's performance or economy.

ACTION

Refer to the following Service Procedure for diagnostics and injector replacement.

SERVICE PROCEDURE

The cackle noise will be found early in an engine's life due to low fuel pressure and/or the relationship of injectors and engine firing order, or anytime there is a drop in fuel pressure. If a mature engine develops the cackle noise, low fuel pressure is the likely cause. Before making a repair attempt, the following tests should be performed:

- Verify that the noise occurs with a hot engine, not cold.
- Verify that the vehicle has been driven at least 32 km (20 miles) since the last injector replacement. Air trapped in the system could cause a noise that sounds like cackle.
- If the fuel pressure is less than 379 kPa (55 psi) in either head, repair it first and re-evaluate the cackle concern.
- Repair any other engine performance concerns prior to attempting a repair for cackle. Example: low pressure concerns may be causing low power output and cackle; repairing the low fuel pressure may eliminate both concerns. Also, an engine miss may be an indicator of valvetrain concerns.
- If you have an Injector Performance Analyzer, determine if the cackle is coming from cylinder #8. Cancel that cylinder to determine whether the noise still exists. Cackle cannot be identified using the Cylinder Contribution Test.

If the noise is hot only, and 1) no performance concern exists, 2) fuel pressure is above 379 kPa (55 psi), 3) the engine or the injector in cylinder #8 is new, and 4) the vehicle has been driven at least 32 km (20 miles) since the last injector replacement, the engine is a candidate for a special Injector (XC3Z-9E527-AARM).

NOTE

THE NEW LONG-LEAD INJECTOR FOR CACKLE NOISE IS USED ONLY IN CYLINDER #8. DO NOT USE IN ANY OTHER CYLINDER DUE TO FUTURE CYLINDER CONTRIBUTION TEST

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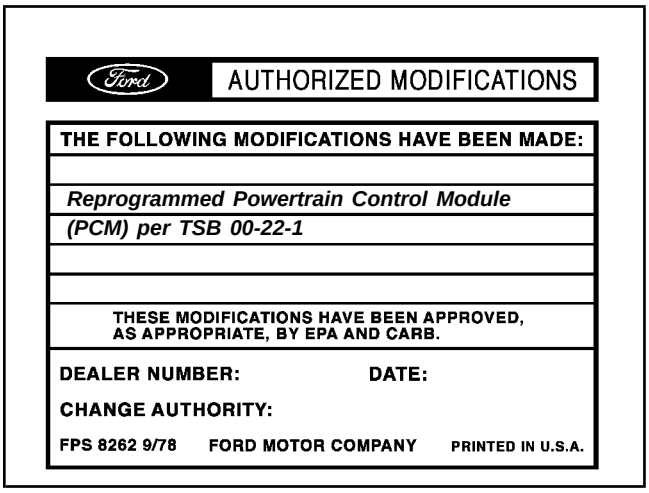
DIAGNOSTICS BEING AT RISK. LONG-LEAD INJECTORS INSTALLED IN ANY CYLINDER OTHER THAN #8 WILL NOT BE COVERED BY FORD WARRANTY. THE INJECTOR IS RECOGNIZABLE BY A BLUE COLORED SOLENOID AND AN "LL" STAMPED ON THE SOLENOID WHERE THE UVC (INJECTOR HARNESS) CONNECTS.

Replace the #8 injector by referring to the replacement procedure in the appropriate Workshop Manual. In addition to replacing the injector, the Powertrain Control Module (PCM) must be reprogrammed to prevent the #8 injector from flagging a false Cylinder Contribution Test code during future diagnostic tests. After the repair, drive the vehicle at least 32 km (20 miles) to purge air from high-pressure oil system (realize that some small amount of cackle may remain).

PCM RECALIBRATION INFORMATION			
Application	Old Tear Tag	New Tear Tag	WDS/NGS Qualifier
F-250/350 M/T Federal Service Fix 1999.5	QEK0	QEK1	9DTM-AN CALIBRATION
F-250/350, F-450/550 M/T Federal Service Fix 1999.5	ZHL0	ZHL1	9DTM-AP CALIBRATION
F-450/550/560 M/T 50S Service Fix 1999.5	VRP0	VRP1	9DTM-AO CALIBRATION
F-450/550/560 M/T 50S Service Fix 1999.5	BAW0	BAW1	9DTM-AQ CALIBRATION
F-250/350 M/T California Service Fix 1999.5	GLI0	GLI1	9DTM-BF CALIBRATION
F-250/350 M/T California Service Fix 1999.5	SBJ0	SBJ1	9DTM-BG CALIBRATION
F-250/350 A/T Federal Service Fix 1999.5	YEV0	YEV1	9DTA-AR CALIBRATION
F-250/350, F-450/550 A/T Federal Service Fix 1999.5	MRT0	MRT1	9DTA-AU CALIBRATION
F-450/550/560 A/T 50S Service Fix 1999.5	RXM0	RXM1	9DTA-AS CALIBRATION
F-450/550/560 A/T 50S Service Fix 1999.5	LVS0	LVS1	9DTA-AV CALIBRATION
F-250/350 A/T California Service Fix 1999.5	XWF0	XWF1	9DTA-BH CALIBRATION
F-250/350 A/T California Service Fix 1999.5	HQN0	HQN1	9DTA-BK CALIBRATION
E-350/450 A/T Federal Service Fix 1999.5	UYQ0	UYQ1	9DTA-AO CALIBRATION
E-350/450 A/T California Service Fix 1999.5	JUK0	JUK1	9DTA-BG CALIBRATION
E-350/450 A/T California Service Fix 2000	CHK3	CHK5	9DTA-BG CALIBRATION
E-350 A/T California Service Fix 2000	MYA0	MYA1	9DTA-BM CALIBRATION
F-250/350 M/T California Service Fix 2000	ATA3	ATA4	9DTM-BG CALIBRATION
F-250/350 M/T California Service Fix 2000	FHP3	FHP4	9DTM-BK CALIBRATION
Excursion A/T 50S Service Fix 2000	ZYS5	ZYS6	0L119Y0B CALIBRATION
Excursion A/T 50S R/C 2000	AXD0	AXD1	0L119Y0B CALIBRATION
F-250/350 M/T Federal R/C 2000	DAC3	XPC4	9DTM-AP CALIBRATION
F-450/550 M/T California R/C 2000	VXY3	VXY4	9DTM-AQ CALIBRATION
F-250/350 A/T California R/C 2000	NVK3	NVK5	9DTA-AU CALIBRATION
F-250/350 A/T California R/C 2000	CNH4	CNH5	9DTA-AV CALIBRATION
F-250/350 M/T California R/C 2000	HMR0	HMR1	0F729M0B CALIBRATION
F-450/550 A/T CFFV R/C 2000	YLE0	YLE1	0F719P0B CALIBRATION
F-250/350 A/T California R/C 2000	UKN0	UKN1	0F719N0B CALIBRATION
E-350/450 A/T Federal R/C 2000	SZT3	SZT5	9DTA-AO CALIBRATION
E-350/450 A/T Federal R/C 2000	RCU0	RCU1	9DTA-AZ CALIBRATION
E-350/450 A/T California R/C 2000	FHM1	FHM2	0E419Q0B CALIBRATION
E-350 A/T California R/C 2000	SBR1	SBR2	0E419R0B CALIBRATION

Obtain an Authorized Modifications Decal (FPS 8262 - obtainable through DOES II, 25/pkg) and list the date, dealer number, and summary of modifications performed. Select a prominent place adjacent to the Vehicle Emission Control Information Decal suitable for installing the Authorized Modifications Decal. Clean the area, install the decal, and cover it with a clear plastic decal shield.



PART NUMBER	PART NAME
XC3Z-9E527-AARM	#8 Injector

Article No. 00-22-1 Cont'd.

OTHER APPLICABLE ARTICLES: NONE
SUPERSEDES: 00-10-1
WARRANTY STATUS: Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage And Emissions Warranty Coverage

OPERATION	DESCRIPTION	TIME
002201A	Replace The #8 Injector (Includes Time To Test Drive And Reprogram PCM) - 1999-2000 Super Duty F Series And 2000 Excursion Models	1.9 Hrs.
002201B	Replace The #8 Injector (Includes Time To Test Drive And Reprogram PCM) - 1999-2000 Econoline	2.9 Hrs.

DEALER CODING

BASIC PART NO.	CONDITION CODE
9E527	07

OASIS CODES: 497000, 702100